ZS-A28

Vacuum Energy as a Projector-Valued Top Form: A Conditional Closure of the Z-Spin Vacuum-Identification Bridge (B3-D)

with the microscopic scale-generation No-Go (Part I) and the X–Y holographic boundary law (Part II) — the absolute vacuum-energy scale B3

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v2.0 Load-bearing audit: 27/27 real fail-closed asserts PASS (attached, independently runnable: verify_zs_a28_v2_0.py) | full corpus regression suite (≈111 checks) maintained separately | **Zero Free Parameters** (modulo the present-epoch normalization, §16.6) | (**A** = 35/437, **Q** = 11, dim **Z** = 2) LOCKED *Status of the central claim. The absolute vacuum-energy scale* **B3** *remains* **OPEN** *(the absolute value is forbidden by the A–Q-Only No-Go and isolated as the central-shift offset c_0 of ZS-A26). B3 is decomposed cleanly: Part I shows every microscopic engine merely re-expresses one number N; Part II reframes B3 as a UV/IR boundary that* **factorizes** *the hierarchy into $\rho_\Lambda^{1/4} = \sqrt{(\bar{M}_P H_0^*)}$* — a localization, not a resolution. For the Vacuum-Identification bridge B3-D — $q_\Lambda \mapsto T_{\mu\nu}^\Lambda = -\rho_\Lambda g_{\mu\nu}$ — *the* ***field-theory half*** *is* ***IMPORTED-PROVEN*** *(a covariant top-form on the rank-83 complement* ***P_Λ*** *forces $w = -1$, $\partial_\mu\rho_\Lambda = 0$, $T_{\mu\nu}^\Lambda = -\rho_\Lambda g_{\mu\nu}$*), and the* **identification half** *is a* **DERIVED-CONDITIONAL** *implication on an explicit set {Vacuum-Support, Top-Form-Selection, Unified-Normalization, Physical-Trace$_{121}$*}.* v2.0 makes two further distinctions the reviews demanded. (1) The* **rank fraction** *$q_\Lambda = \tau_{121}$*$(P_\Lambda$*) = 83/121 (a projector fact, closed modulo the decidable 121-vs-120 trace gate) is separated from the* **energy fraction** *$\Omega_{\Lambda,0}$ = 83/121 (today's cosmological observable); the parameter-free status of the flagship number rests on the present-epoch normalization. (2)* **Unified-Normalization cannot be a local identity** *— matter dilutes ($\rho_m \propto a^{-3}$) while the vacuum is constant — so it is re-cast as a present-epoch / charge–flux selection, i.e. the cosmological coincidence problem, which is the true bridge to B3-C. As an unconditional closure B3-D is* **PARTIAL**; *the crux (the present-epoch 38:83 selection) is handed to* **ZS-A29** *with an explicit program.*

Corrections, resolution, unification, reframe, repair of record, and the B3-D conditional closure (v1.0 → v2.0)

v2.0 — RANK vs ENERGY FRACTION SEPARATED; UN RE-CAST AS A PRESENT-EPOCH SELECTION; A29 SPUN OUT. v2.0 integrates two further independent reviews of v1.9 and makes three substantive fixes plus editorial repairs. (i) §12 split (mandatory fix). The rank fraction $q_\Lambda = \tau_{121}(P_\Lambda)$ = 83/121 (a projector fact, PROVEN modulo the physical-trace denominator PT) is now cleanly separated from the energy fraction $\Omega_{\Lambda,0} = \rho_{\Lambda,0}/\rho_{tot,0}$ = 83/121 (today's observable, DERIVED-CONDITIONAL on {VS, TFS, UN, PT, Frame}). Earlier drafts wrote "the corpus already derives Ω_Λ" and "the coefficient is not free," which conflated the two; §12 is retitled and rewritten. The parameter-free status of the flagship number rests on UN — stated openly. (ii) UN time-dependence (substantive physics). The v1.9 local identity ZF $f^2 = 2\ \mu_Z\ n$ is impossible: dust dilutes ($n \propto a^{-3}$) while the top-form is constant, so the equality holds on at most one hypersurface (verify block D, E). UN is re-cast as a present-epoch / global charge–flux selection (§16.6) — the cosmological coincidence problem in Z-Spin form, and the genuine bridge to B3-C. Gate XB-6b renamed Present-Epoch Boundary Charge–Flux Matching. (iii) Structural dilemma made explicit (§16.3): the top-form vacuum (which gives $w = -1$) and single-trace protection of 38:83 are mutually exclusive — a separate carrier is exactly what frees the ratio — so v1.9's demotion was forced, not optional (verify block G). (iv) "dissolves the hierarchy" → "factorizes / localizes" (§11, §15). (v) Verification record fixed: the attached script now uses 27 real fail-closed asserts (verify_zs_a28_v2_0.py); the cover states this load-bearing count, not a single "111/111," and flags the inherited corpus suite as maintained separately. (vi) Residual v1.8 tag fixed (§15). (vii) Crux spun out to ZS-A29 (new §17): A Single-Parent Measure/Form Action for Dust and Vacuum — Global Charge–Flux Matching, the Present-Epoch Selection Problem, and the 38:83 No-Go or Closure, with a Guendelman–Nissimov–Pacheva single-measure route [27,28] and a BV–BFV [29] PT computation. Net: A28 reaches its in-paper closure ceiling; field-theory half IMPORTED-PROVEN, identification half DERIVED-CONDITIONAL, unconditional closure PARTIAL, absolute scale terminal in B3-B. New references [27–29]; updated §11, §12, §15, §16.3/§16.6/§16.7, gates, Appendix D (Pass C).

B3-D RECLASSIFIED — FIELD-THEORY HALF IMPORTED-PROVEN, IDENTIFICATION HALF DERIVED-CONDITIONAL ON FOUR NAMED ANTECEDENTS (v1.9). v1.9 audits the v1.8 B3-D closure against two independent adversarial reviews and corrects a load-bearing overclaim. (i) Single-trace scope corrected. v1.8 claimed the ZS-A20 §9.3 single-trace structure protects the 83:38 matter-to-vacuum ratio against reabsorption (status DERIVED). This is wrong: single-trace fixes ratios only within one carrier. Baryon and CDM share the same Brown current (one coefficient μZ), so 6:32 is protected (DERIVED). The vacuum is a separate top-form with its own coupling (Z_F, f), so the cross-carrier ratio carries a free factor $(Z_F f^2)/(2\mu Z n)$; 38:83 is OPEN, promotable to DERIVED-CONDITIONAL only under an explicit Unified-Normalization condition (verify block C; the audit Python is reproduced in Appendix D). (ii) "Single remaining debt" retracted. B3-D does not reduce to one condition; its residual is the set {Vacuum-Support, Top-Form-Selection, Unified-Normalization, Physical-Trace$_{121}$}. The Vacuum-Support Lemma ($s(J_{matter}) = P_b + P_c$) does not imply the Top-Form-Selection Lemma ($\mathrm{supp}\, F_4 = P_\Lambda$): "no matter current here" does not entail "a top-form lives here." (iii) Vacuum-Support status corrected from "DERIVED-CONDITIONAL on ZS-U4" to DERIVED-CONDITIONAL on an explicit Exhaustive-Matter-Content axiom — U4's adoption of the (6,32,83)/121 budget does not prove action-level support-exhaustion. (iv) Symmetry-uniqueness overclaim removed (§16.2): $T \propto g$ is the metric variation of the chosen Maxwell four-form action, not a general consequence of "no preferred direction." (v) The unified action is now written explicitly (§16.4): $S = S_{grav} + S_{Brown}[J_\mu, P_b, P_c] + S_F[F_4, P_\Lambda] + S_{constraint}$, which is what exposes the free cross-carrier coupling. Net: the field-theory half stays solid (IMPORTED-PROVEN); as an unconditional closure B3-D is PARTIAL. The absolute scale remains in B3-B (OPEN). Record fixes: title lightened; M_{111} → M_{121} typo; cover date March → June 2026; "preserved verbatim" → "summarized; incorporated by reference."

GROUNDED (v1.6). 83 is DERIVED in the corpus, not introduced by A28: ZS-F2 §11.4 face-counting ($\Omega_b = XZ/Q^2 = 6/121$, Ω_{cdm} = F(truncated icosahedron)/Q^2 = 32/121, cross-check $XQ-1 = 32$) → ZS-U4 (locked late-time budget) → ZS-A19 (rank-32 conserved dust) → ZS-A20 ($\rho_b/\rho_c = 6/32$ at action level). The dark-energy entry $83 = 121 - 6 - 32 = 82 + 1 = Q^2 - X(Z+Q) + \mathrm{ind}-(DZ)$: a trace complement, and the assignment direction is theorem-forced (baryon = X-sector current → 6; CDM = unique Ih mediator

→ 32), so $c^2 = 83/121$ is DERIVED-COND — not a hidden fit. XB-3 retired-as-inapplicable: corpus DE is constant Λ ($w = -1$, ZS-U1/U4) and the holographic scaling is static (de Sitter entropy, A26/A27) — no Hubble-vs-event-horizon choice; the dynamical $w(z)$ detour is dropped, resolving the U4 conflict. ΔNeff corrected: $\Delta N_{\mathrm{eff}}^{\mathrm{BBN}} = 0.160$ survives (D/H), but always-on $\Delta N_{\mathrm{eff}}^{\mathrm{CMB}} = 0.160$ is FALSIFIED ($\Delta\chi^2 = +408$); the clean channels are $\Delta N_{\mathrm{eff}}^{\mathrm{BBN}}$ and $r = 0.0089$. Theorem A28.10 is now a conditional implication, not an iff. §0, §12–§15. REPAIRED (v1.5). The v1.4 unification Theorem A28.10 conflated two different objects: $c^2 = \Omega_\Lambda = 83/121$ is a $Q^2 = 121$ face budget (the 9-block tensor $(3,2,6)^{\otimes}(3,2,6)$, with Λ the remainder 83), whereas Condition C is the $Q = 11$ sector weight $(3,2,6)/11$ — and no sector weight equals 83/121. v1.5 decouples them. REFRAMED (v1.4). Part I (the microscopic route) is preserved as an audit; Part II reframes B3 as a UV/IR boundary problem. UNIFIED (v1.3) / RESOLVED (v1.2) / RETIRED: both internal engines re-express N; the chiral-Pfaffian ½ is not forced; the spectral route is retired. RETRACTED (v1.0 §7): non-unitary torsion. DEMOTED: v1.0 Thms A28.2/A28.4/A28.3/A28.5; B3-4 PARTIAL. PRESERVED: Thm A28.1 (mass-dimension correction) PROVEN.

§0. Abstract

ZS-A27 reduced the absolute vacuum-energy scale B3 to a four-root programme; ZS-A28 v1.0–v1.3 attacked the microscopic sequestering trunk and proved, blind, that the internal scale-generation route is **closed-negative** — the chiral Pfaffian ½ is fitting, not forced, and swapping the two internal engines merely relabels one number $N = \ln(\bar{M}_P/\rho_\Lambda^{1/4}) \approx 69.16$ without closing it. Part I is preserved as that audit. **Part II reframes the problem:** vacuum energy is a UV/IR boundary quantity, not a microscopic constant. $N = \tfrac{1}{2}\ln(\bar{M}_P/H_0) - \tfrac{1}{4}\ln(3c^2)$, so $\rho_\Lambda^{1/4} = \sqrt{\bar{M}_P H_0}$ is a geometric mean and the ~120-order tuning *factorizes* into one O(1) rank coefficient $\mathbf{q}_\Lambda = 83/121$ plus a residual absolute scale (which stays in B3-B) — a localization of the naturalness problem, not a resolution.

83 is DERIVED in the corpus, not introduced here. ZS-F2 §11.4 face-counting gives $\Omega_b = XZ/Q^2 = 6/121$ (baryon = X-sector matter with a Z_2 seam charge, **DERIVED**) and Ω_{cdm} = F(truncated icosahedron)/Q^2 = 32/121 (CDM = the *unique* Ih mediator; cross-checked $XQ-1 = 32$); ZS-U4 adopts the budget (6,32,83)/121 as the locked late-time input; ZS-A19/A20 realise 32 as a conserved rank-32 boundary-tension dust and derive $\rho_b/\rho_c = 6/32 = 3/16$ at action level. The dark-energy entry **83 = 121 − 6 − 32** is therefore the trace *complement* of the 38 matter modes. The assignment direction (visible = 6, dark = 32) is *theorem-forced*, so the **rank fraction** $\mathbf{q}_\Lambda = 83/121$ is **DERIVED-COND** (modulo the 121-vs-120 trace gate) — not a hidden fit. v2.0 carefully distinguishes this *rank* fraction from the *energy* fraction $\boldsymbol{\Omega}_{\Lambda,0} = \rho_{\Lambda,0}/\rho_{tot,0}$: the two coincide only at the present epoch (matter dilutes, the vacuum does not), so $\boldsymbol{\Omega}_{\Lambda,0} = 83/121$ is a conditional, not unconditional, statement (see below).

v2.0 establishes the field-theory half of the Vacuum-Identification bridge B3-D, reclassifies the identification half as a conditional implication, and separates the rank fraction from the energy fraction. The **field-theory half** is **IMPORTED-PROVEN**: realizing the rank-83 trace complement $\mathbf{P}_\Lambda = \mathbf{I}_{121} - \mathbf{P}_b - \mathbf{P}_c$ as the internal support of a covariant **top-form** ($\mathbf{F}_4 = d\mathbf{A}_3$), the PROVEN top-form theorems [23–26] *force* no local propagating degrees of freedom, on-shell constancy ($\partial_\mu\rho_\Lambda = 0$), and **T**μν**^Λ = −ρ**Λ**g**μν, i.e. $\mathbf{w = -1}$. The **identification half** is a **DERIVED-CONDITIONAL**

implication on {**Vacuum-Support**, **Top-Form-Selection**, **Unified-Normalization**, **Physical-Trace**$_{121}$}. Two corrections, carried from v1.9 and sharpened in v2.0: (a) the ZS-A20 §9.3 single-trace structure protects only the *within-matter* ratio $\rho_b : \rho_c$ = **6 : 32** (one Brown current, one coefficient μ_Z), **not** the *cross-carrier* ratio $\rho_{matter} : \rho_\Lambda$ = **38 : 83** — the vacuum is a *separate* top-form with a free relative normalization $(Z_F f^2)/(2\mu_Z n)$; and this is *forced*, because the very feature that gives $w = -1$ (a separate carrier) is what frees the ratio (the structural dilemma, §16.3). (b) v2.0 shows the would-be Unified-Normalization $Z_F f^2 = 2\mu_Z n$ **cannot be a local identity** — dust dilutes ($\rho_m \propto a^{-3}$) while the top-form is constant — so it is re-cast as a **present-epoch / charge–flux selection** (§16.6), the cosmological coincidence problem, and the genuine bridge to B3-C. Consequently the *rank* fraction $\mathbf{q}_\Lambda$ = 83/121 is closed (modulo PT) while the *energy* fraction $\mathbf{\Omega}_{\Lambda,0}$ = 83/121 is **DERIVED-CONDITIONAL on {VS, TFS, UN, PT, Frame}** — the parameter-free status of the flagship number rests on the present-epoch normalization, stated openly. The **absolute** normalization stays **reassigned to B3-B** (ZS-A19.NG2, ZS-A26). As an *unconditional* closure of B3-D the status is **PARTIAL**; the crux (the present-epoch 38:83 selection, a single-parent action that might derive VS/TFS/UN jointly) is handed to **ZS-A29** (§17).

The precise B3 classification (v2.0). B3-A (A–Q-only absolute) **CLOSED-NEGATIVE**; B3-B (with an extra dimensionful datum, ≡ the central-shift offset c_0) **OPEN**; B3-C (today's relation) **DERIVED-COND**; B3-D (the vacuum-identification bridge) — field-theory half **IMPORTED-PROVEN**, identification half **DERIVED-CONDITIONAL** on {VS, TFS, UN, PT}, hence **PARTIAL** as an unconditional closure — with the *rank* fraction $\mathbf{q}_\Lambda$ = 83/121 closed (mod PT) and the *energy* fraction $\mathbf{\Omega}_{\Lambda,0}$ = 83/121 DERIVED-CONDITIONAL on the present-epoch normalization. So B3 is *reframed, factorized, localized, and its identification bridge conditionally derived*; the load-bearing OPEN items are the four B3-D antecedents — chief among them the present-epoch charge–flux selection (XB-6b → ZS-A29) — and, terminally, the absolute scale in B3-B.

Predictions. The clean falsifiable channels are $\Delta N_{eff}^{BBN} = \dim(Z)\cdot\mathbf{A} = 2\mathbf{A} = 0.160$ (resolving the D/H tension to −0.05σ) and $r = 0.0089$ (tensor-to-scalar; high-significance at LiteBIRD nominal sensitivity, $\delta r < 0.001$). The *always-on* $\Delta N_{eff}^{CMB} = 0.160$ scenario is **FALSIFIED** by the corpus Cobaya analysis ($\Delta\chi^2_{CMB} = +408$). The dimensionless $\mathbf{\Omega}_{\Lambda,0}$ = 83/121 ≈ 0.686 matches observation but is parameter-free only under the present-epoch normalization (above). The attached **verify_zs_a28_v2_0.py** runs **27/27** load-bearing checks fail-closed (real asserts); the ≈111-check corpus regression suite is maintained separately. (**A**, **Q**, dim **Z**) = (35/437, 11, 2) **LOCKED**.

Epistemic Status Legend

PROVEN — established by internal computation/algebra at machine precision.
IMPORTED-PROVEN — proven in the external literature and used here without re-proof; full citation given.
DERIVED — follows from the Z-Spin action plus PROVEN inputs, zero free parameters.
DERIVED-CONDITIONAL — DERIVED given explicitly named external/corpus conditions.
HYPOTHESIS — well-motivated, pre-registered, not yet derived; falsifiable.

PARTIAL — a closure path is specified and partly executed; the load-bearing computation is pending.
TARGET-COMPUTATION — a finite, decidable computation set up but deferred.
OPEN — not closed by current corpus tools; no committed probability.
CLOSED-NEGATIVE — a route proven not to work (a no-go).
RESOLVED / REPAIRED / REFRAMED / RETIRED / RETAINED / DECOUPLED — version-management tags.
RETRACTED / DEMOTED — a claim withdrawn or lowered in status by audit

§1. Introduction: from three weak handles to physical objects

ZS-A27 §7 proved the **A–Q-Only Scale-Generation No-Go**: the locked dimensionless data cannot fix a dimensionful $\rho\Lambda$. This is not a wall — it says only that *A* and *Q* alone are insufficient; closing B3 with the one locked dimensionful anchor $\bar{M}P$ plus a dynamical scale-generating mechanism does not violate it. ZS-A28 v1.0 attacked the sequestering trunk in the order ZS-A27 §8.2 prescribed and found a real dimensional error in A27 (the source term $\bar{M}P^2 \to \bar{M}P^{-2}$). But an external audit showed v1.0's trunk rested on three weak handles: a Wilson-LOCATOR Gram matrix standing in for the kernel, a *declared* membrane branch standing in for branch selection, and a torsion route mis-labelled 'non-unitary'. v1.1 retracts these and rebuilds on physical objects.

The reorganisation is driven by one observation: the abstract odd operators Ωa of v1.0 have two canonical, physical realizations in four dimensions — the **gauge** and **gravitational Pontryagin densities**,

$$\Omega 1- = (1/32\pi^2)\ \mathrm{tr}\ F- \wedge F-, \quad \Omega 2- = (1/384\pi^2)\ R \wedge R, \qquad \text{(A28.30)}$$

both pseudoscalar (JZ-odd, since γ^5 = JZ and the chiral anomaly is $\partial\mu j5\mu \propto \mathrm{tr}\ F\tilde{F}$), gauge/diffeomorphism invariant, dimension 4, and expressible as four-forms with well-defined *topological susceptibility*. This single substitution makes B3-2, B3-4, and B3-5 the same physics, and is what the rest of Part I develops.

[Part I (§§2–9) — the microscopic-engine audit — is **summarized from v1.7**; the detailed derivations are **incorporated by reference** (v1.7 §§2–9). The substantive content is unchanged; v1.9 adds no new physics to Part I, and reproduces only its results in condensed gate form below. Readers requiring the full step-by-step derivations should consult v1.7 §§2–9, on which this summary depends.]

§2.–§9. Part I (summarized from v1.7; full derivations incorporated by reference): why microscopic scale-generation fails

The one v1.0 result that survives audit unchanged is **Theorem A28.1** (the dimensional correction of record): with $[qa] = 2$, $[\Omega a] = 4$, the JZ-odd doublet action $S- = \int d^4x \sqrt{-g}\ [-\tfrac{1}{2}\ qa(Kq^{-1})ab\,qb + \bar{M}P^{-2}\ qa\Omega a]$ gives $\rho M1 = (2\bar{M}P^4)^{-1}\ \Omega^T Kq\Omega \geq 0 \Leftrightarrow Kq \geq 0$ (**PROVEN**). With $\chi Z = -1$ a single four-form gives $\rho < 0$, so the doublet is mandatory.

Theorem A28.7 (Symplectic Selection, DERIVED-CONDITIONAL). The odd cohomology $HD^{-} \cong \mathbb{C}^2$ (Kostant, ZS-M27) carries a Kramers structure $C^2 = -1$ (spin-½, ZS-Q14); the connected group preserving h and $\omega(u,v) = h(Cu,v)$ is Aut(h,ω) = USp(2) ≅ **SU(2)**Z. Pure SU(2)Z has $b_0 = 22/3 > 0$ and no fermion doublets, so the Witten anomaly is absent. **Theorem A28.8 (Susceptibility Kernel, DERIVED-CONDITIONAL).** $\chi ab = V_4^{-1}\langle QaQb\rangle c$ is PSD by construction; rank ≥ 1 iff $\chi_{11} > 0$ (the positive Yang–Mills topological susceptibility). B3-2 (rank) and B3-5 (coefficient) are one quantity. **Theorem A28.9 (Chiral Pfaffian ½: structure PROVEN; assignment HYPOTHESIS).** Pf^2 = det gives a structural ½, but its assignment to ln 3 rather than ln 2 is *not* forced (the real subfield of $\mathbb{Q}(\sqrt{-3},\sqrt{-11})$ is $\mathbb{Q}(\sqrt{33})$, conductor 33). The internal spectral route is therefore **RESOLVED closed-negative**.

The hierarchy-exponent unification (§7.5). The absolute-scale problem is one number, the hierarchy exponent $N \equiv \ln(\bar{\mathbf{M}}\mathrm{P}/\rho\Lambda^{1/4}) = 69.16$, and both internal engines are two parametrizations of it: Engine B (CW/Pfaffian) $N = 2\gamma\mathrm{CW}C\mathrm{oddsp}$; Engine A (SU(2)Z transmutation) $N = 16\pi^2/(b_0 g\mathrm{Z}^2)$. At their *forced* values both engines miss, and each matches only via the sealed target. **The engine swap does not close S1 — it relabels the same gap.** This is the sharpest honest statement of Part I: the cosmological-constant hierarchy is one undetermined number that dimensionless invariants cannot fix. **B3 remains OPEN** at the end of Part I, neither closed nor proven impossible.

——— **PART II: THE X–Y VACUUM BOUNDARY LAW** ———

§10. The reframe: vacuum energy is a UV/IR boundary, not a microscopic constant

Part I converged on a single negative fact: every microscopic engine merely re-expresses the one number $N = 69.16$ without predicting it, because the A–Q-Only No-Go forbids generating a large dimensionless hierarchy from locked dimensionless data. The resolution is not a better engine but a *category correction*. Quantum field theory computes the vacuum as the zero-point energy summed to the UV cutoff (~ $\bar{\mathbf{M}}\mathrm{P}^4$, a Y-sector quantity); what is observed is the faint IR floor that an extended X-sector region can carry. The 120-order 'discrepancy' compares a Y-ceiling to an X-floor: a mismatch of layers, exactly the X / Y separation the corpus posits.

Made precise, this is a *UV/IR relation*. The decisive identity (verify block H) is

$$N = \ln(\bar{\mathbf{M}}\mathrm{P}/\rho\Lambda^{1/4}) = \tfrac{1}{2}\ln(\bar{\mathbf{M}}\mathrm{P}/H0) - \tfrac{1}{4}\ln(3c^2), \qquad \text{(A28.43)}$$

so $N = 69.16$ is simply *half the logarithmic distance between the Planck and Hubble scales* (up to an O(1) term). Equivalently

$$\rho\Lambda^{1/4} = (3c^2)^{1/4}\sqrt{\bar{\mathbf{M}}\mathrm{P}H0} \quad \text{— the geometric mean of the UV ceiling and the IR floor.} \qquad \text{(A28.44)}$$

This **reframes and localizes** the hierarchy: N is a ratio of two physical scales, which dimensionless invariants are not asked to fix, and the fine-tuning *factorizes* into a single **O(1)** number $c^2 \approx 0.686$ — the kind of object **A** = 35/437 and **Q** = 11 can produce — together with the residual absolute scale that remains in B3-B. This is a *factorization* of the naturalness problem, not a resolution of it: the radiative-stability and absolute-normalization parts are localized, not solved. The danger to guard against is *import-and-relabel*: holographic dark energy already exists, so the corpus earns its keep only if c^2 follows from **A**, **Q**.

§11. The X–Y vacuum boundary law and its external lift

Let an X-sector region have radius L. The Cohen–Kaplan–Nelson bound [15] gives $\rho\Lambda \lesssim \bar{M}P^2/L^2$; saturating at an IR horizon (Li normalization [16]), $\rho\Lambda = 3c^2\bar{M}P^2/L^2$ (A28.45). With $L = 1/H$ and Friedmann $3\bar{M}P^2H^2 = \rho tot$, $c^2 = \Omega\Lambda$. **External lift (frame H — static de Sitter entropy).** The CKN UV/IR bound [15] supplies the motivation; the Gibbons–Hawking/Wald de Sitter entropy $SdS = AdS/4G = 8\pi^2\bar{M}P^2/H\infty^2$ [18] (carried by ZS-A23, ZS-U1) supplies the mechanism, fixing $\rho\Lambda = 3\bar{M}P^2H\infty^2$ at the attractor and $\rho\Lambda/\rho crit = \Omega\Lambda{,}0 = 83/121$ today. So $\rho\Lambda \sim \bar{M}P^2H^2$ is **DERIVED** (static de Sitter entropy); what remains Z-Spin-specific is the value $\Omega\Lambda{,}0 = 83/121$.

§12. The rank coefficient qΛ = 83/121 and its conditional promotion to the energy fraction ΩΛ,0

A distinction that v1.9 left implicit and v2.0 makes explicit governs this whole section. There are **two different objects** that both evaluate to 83/121, and only one of them is closed:

- the **rank (trace) fraction** $\mathbf{q}\Lambda \equiv \tau_{121}(\mathbf{P}\Lambda)$ = rank $\mathbf{P}\Lambda$ / Q^2 = 83/121, a *finite-algebra fact* about the projector, independent of cosmic time; and
- the **energy fraction** $\mathbf{\Omega}\Lambda{,}0 \equiv \rho\Lambda{,}0 / \rho tot{,}0$, a *cosmological observable* about today's universe.

These coincide **only** under the present-epoch normalization of §16 (Condition UN); in general they differ, because matter dilutes ($\rho m \propto a^{-3}$) while the vacuum is constant ($\rho\Lambda$ = const), so the energy fraction is time-dependent while the rank fraction is not. The corpus closes **q**Λ; it promotes **q**Λ to **Ω**Λ,0 only conditionally. Of Q^2 = 121 boundary slots, 6 + 32 are matter and **83** remain for the vacuum complement, giving

qΛ = rank **P**Λ / 121 = 83/121 = 0.6860, **q**Λ^(1/2) = 0.8282. (A28.46)

Table 12.1. The (6, 32, 83)/121 *rank* budget vs the observed *energy* fractions.

Quantity	Corpus rank value	Observed energy fraction	Status
baryon rank / Q^2	6/121 = 0.0496	$\Omega b \approx 0.0493$	**q-fraction DERIVED-COND; energy match conditional**
CDM rank / Q^2	32/121 = 0.2645	$\Omega cdm \approx 0.265$	**q-fraction DERIVED-COND; energy match conditional**
vacuum rank / $Q^2 = q\Lambda$	83/121 = 0.6860	$\Omega\Lambda,0 \approx 0.685$	**qΛ PROVEN (mod PT); ΩΛ,0 DERIVED-COND on VS+TFS+UN+PT+Frame**
ρΛ relation	$3\ q\Lambda\ \bar{M}P^2H_0^2$	pred/obs ≈ 1.01	**DERIVED-COND (uses H_0)**
rank ratio 83:38	83/38 = 2.184	2e^A = 2.167 (0.8%)	**rank fact; energy ratio conditional on UN**

The corpus derivation lineage (of the rank fraction). 83 descends through four corpus papers: **ZS-F2 §11.4** (6/121, 32/121, cross-check XQ−1 = 32) → **ZS-U4** (locked late-time budget) → **ZS-A19** (rank-32 conserved Brown–Kuchař dust, an equivariant rank-32 projector in the 121-channel algebra) → **ZS-A20** (ρb/ρc = Tr Pb/Tr Pc = 6/32 from a single-trace matrix current at the action level). The assignment direction is theorem-forced (CDM = 32 a polyhedral theorem; baryon = 6 the softer ZS-F5 B3.1 leg), so the *rank* fraction **q**Λ = 83/121 is **PROVEN as a trace fact, DERIVED-CONDITIONAL on the physical-trace denominator (PT)**. What this lineage does **not** establish on its own is that the *energy* fraction equals the rank fraction today: that is the **Vacuum-Identification + present-epoch-normalization** content of §16, and it carries the status **Ω**Λ,0 = 83/121 = **DERIVED-CONDITIONAL on {VS, TFS, UN, PT, Frame}**.

Honest framing of the flagship number. Because the corpus's headline cosmological prediction is the dimensionless **Ω**Λ,0 ≈ 0.686, this distinction is load-bearing: the "zero-parameter" status of **Ω**Λ,0 = 83/121 *rests on* the present-epoch normalization UN (§16.6). If UN is derived (gate XB-6b / ZS-A29), 83/121 survives as a parameter-free energy fraction; if UN turns out to be a genuine free input, 83/121 remains an exact statement about the *rank* but becomes a *fit* at the level of the *energy* fraction. v2.0 registers this openly rather than asserting the energy fraction unconditionally as earlier drafts did.

The 121-vs-120 denominator — the Physical-Trace gate PT (OPEN, decidable). The +1 in 83 = 82 + 1 labels the Z_2-odd mode removed from CDM (33 → 32). Case A (the removed mode is a physical background mode; denominator $Q^2 = 121$, **q**Λ = 83/121 = 0.6860) vs Case B (a pure-gauge mode fully removed; denominator 120, **q**Λ = 82/120 = 0.6833) is registered **OPEN** but **decidable** by a BRST / BV–BFV physical-state cohomology computation (gate PT; cf. ZS-A29 session 3). Numerically both lie within ≈ 1% of the observed ΩΛ ≈ 0.6847 (|0.6860 − obs| ≈ 0.0013; |obs − 0.6833| ≈ 0.0014), so present data does *not* discriminate them; the resolution is theoretical. PT shifts the *value* (83/121 ↔ 82/120) but, as §16 shows, *not* the $w = -1$ vacuum structure.

§13. The unification theorem (repaired): B3 = P0 = [face-budget c²], with Condition C decoupled

Theorem A28.10 (Holographic reduction — conditional implication, DERIVED-CONDITIONAL on frame H). With $\rho\Lambda = c^2 \cdot 3\bar{\mathbf{M}}P^2 H^2$: (i) **B3** is fixed once c^2 is known; (ii) at the Hubble/de-Sitter cutoff Friedmann gives $c^2 = \Omega\Lambda$; (iii) $\Omega\Lambda$ = **83/121**, the *face budget*. Hence the *conditional implication* **[Frame H + Φface + horizon-energy source coupling] ⇒ B3-relational**. This is **not** an iff: $c^2 > 0$ by itself does not manufacture a positive cosmological source at the action level — and *that action-level step is exactly what §16 supplies via the top-form*. The two corpus weight triples (3,2,6)/11 (Hilbert register) and (3,3,5)/11 (operator algebra AZS = $M_3 \oplus \mathbb{C} \oplus M_5$) are distinct objects (A19.NG1 resolved), and c^2 = 83/121 lives at the *Q^2-face* level, decoupled from the Q-level Condition C.

The closure gates (updated in v1.9). XB-1 the X–Y boundary theorem (derive $\rho X V X = c^2 E$maxY from a horizon Hamiltonian, **OPEN**); **XB-2** the coefficient c^2 = 83/121 — **DERIVED-COND** (theorem-forced assignment, equal-weight Φface state-sum residual); **XB-3** causal-scale uniqueness — **RETIRED** as inapplicable (constant Λ removes the dynamical-cutoff problem); **XB-4** dynamics $w = -1$ (constant Λ), testable against DESI; **XB-5** the predictions ΔNeffBBN = 0.160 and r = 0.0089. **XB-6 (the B3-D identification half, opened in v1.9, refined in v2.0)** decomposes into: **XB-6a** the **Vacuum-Support Lemma** s(**J**matter) = **P**b + **P**c, via a 121-channel first-order parent action (the exact analogue of ZS-A20 §9.2's ΣZ→Brown derivation); **XB-6b** the **Present-Epoch Boundary Charge–Flux Matching** (formerly "Unified-Normalization") — since the *local* identity ZF f^2 = 2 μZ $n(t)$ is impossible (matter dilutes ∝ a^{-3}, the vacuum is constant; §16.6), the gate is to derive the present-epoch hypersurface Σ_0 on which the energy fraction equals the rank fraction 83/121, from a global relation among matter charge, top-form flux, and boundary three-volume (the cosmological coincidence problem in Z-Spin form; this is the gate that would restore the *energy* fraction **ΩΛ,0** = 83/121 to DERIVED, and the central subject of ZS-A29); **XB-6c** the **Top-Form-Selection Lemma** supp $\mathbf{F}_4$ = **P**Λ, i.e. that the field on the current-null corner *is* a top-form and not, e.g., a constant scalar potential. All three are **OPEN** with explicit symbolic-computation paths; XB-6b is the most consequential for the parameter-free status of the flagship prediction.

§14. The dynamics: constant Λ, the de Sitter entropy, and the clean predictions

The corpus dark energy is constant Λ (w = −1), and that resolves the cutoff. The corpus's actual dark energy is the de Sitter attractor with $w = -1$ *exactly* (ZS-U1 §9, ZS-U4), so ρΛ is constant. The holographic scaling $\rho\Lambda \propto \bar{\mathbf{M}}P^2 H^2$ arises *statically* from the de Sitter horizon entropy (the A26/A27 dilation-anomaly capstone). The present-day relation and the asymptotic de Sitter relation are DISTINCT:

$\rho\Lambda = 3\,\Omega\Lambda{,}0\,\bar{\mathbf{M}}P^2\,H0^2$,   ΩΛ,0 = 83/121   (TODAY, matter present),   (A28.48a)

$\rho\Lambda = 3\ \bar{M}P^2\ H\infty^2 = 24\pi^2\ \bar{M}P^4\ /\ S\mathrm{dS},\infty,\quad H\infty = \sqrt{(83/121)}\ H0$ (ASYMPTOTIC de Sitter). (A28.48b)

Because $\rho\Lambda$ is constant, $\Omega\Lambda(t) = \rho\Lambda/(3\bar{M}P^2H^2)$ varies: today 83/121, at the future attractor → 1. **So 83/121 is today's fraction ΩΛ,0****; the de Sitter saturation coefficient is 1** — different objects. At the attractor the Hubble, apparent, and event horizons coincide (all = $1/H\infty$), so there is no horizon choice to make (XB-3 retired). B3-absolute — predicting $H\infty$ or $H0$ from **A**, **Q** — remains a No-Go.

The clean, scale-independent predictions. ΔNeffBBN = dim(Z)·**A** = 2**A** = 70/437 = 0.1602 (moves D/H from −2.3σ to −0.05σ); the always-on ΔNeffCMB = 0.160 is **FALSIFIED** ($\Delta\chi^2$CMB = +408.27). The clean near-term CMB prediction is $r = 0.0089$ (2.7× Starobinsky; high-significance at LiteBIRD nominal sensitivity), together with $\eta B = (6/11)^{35} = 6.12\times10^{-10}$ (0.02σ) and $\tau p = t\mathrm{P}\ \exp(5\pi/\mathbf{A}) \approx 2.56\times10^{34}$ yr (Hyper-K).

§15. Conclusion: from a failed engine to a falsifiable boundary law with a conditionally-derived vacuum bridge

This paper has three movements. **Part I** audited the microscopic route and converged on a negative result: every scale engine merely re-expresses one undetermined number $N = 69.16$, which the A–Q-Only No-Go forbids deriving. **Part II** reframes the problem along the X / Y duality and *factorizes* the hierarchy: $\rho\Lambda^{1/4} = \sqrt{(\bar{M}PH0)}$, so the ~120-order tuning *localizes* into one O(1) rank coefficient **q**Λ = 83/121 (whose integer 32 is a polyhedral theorem) plus a residual absolute scale that stays in B3-B. **v2.0 (§16)** treats the one bridge that remained — the Vacuum-Identification B3-D — in two halves: it establishes the **field-theory half** as IMPORTED-PROVEN (a covariant top-form on the rank-83 complement *forces* $w = -1$, constancy, and T**μν** = −ρΛ**g**μν), and reclassifies the **identification half** as a DERIVED-CONDITIONAL implication on four named antecedents, carefully separating the *rank* fraction **q**Λ = 83/121 (closed, modulo the physical-trace gate) from the *energy* fraction **Ω**Λ,0 = 83/121 (conditional on the present-epoch normalization), and writing the unified action explicitly.

The honest balance, stated precisely. **B3-relational** — $\rho\Lambda = c^2\cdot 3\bar{M}P^2H^2$ with $c^2 = 83/121$ — is **DERIVED-CONDITIONAL**. **B3-absolute** — predicting ρΛ or $H0$ from **A**, **Q**, **M̄**P alone — is a *non-claim*, forbidden by the No-Go and isolated in v1.8 as the central-shift offset c_0 (ZS-A26). The dynamics are constant Λ ($w = -1$), mildly tensioned by DESI's dynamical-DE preference if it hardens; the clean falsifiable channels are ΔNeffBBN = 0.160 and $r = 0.0089$.

The precise B3 classification (v2.0). B3 splits four ways:
B3-A — A–Q-only absolute generation, (A, Q, M̄P) → H0. CLOSED-NEGATIVE (PROVEN No-Go).
B3-B — absolute scale allowing one extra dimensionful cosmological-state datum ≡ the central-shift offset c_0 (ZS-A26). OPEN — the terminal load-bearing absolute item. B3-C — the present-day relation $\rho\Lambda,0 = 3\ (83/121)\ \bar{M}P^2\ H0^2$. DERIVED-CONDITIONAL (uses H0 — equivalently the present-epoch Σ_0 clock — as input). B3-D — the Vacuum-Identification bridge $q\Lambda \mapsto T\mu\nu\Lambda = -\rho\Lambda g\mu\nu$. Field-theory half IMPORTED-PROVEN (w = −1, constancy, T = −ρg via the top-form theorems [23–26]); identification

half DERIVED-CONDITIONAL on {VS, TFS, UN, PT}; hence PARTIAL as an unconditional closure. The rank fraction $q\Lambda = 83/121$ is closed (mod PT); the energy fraction $\Omega\Lambda,0 = 83/121$ is DERIVED-CONDITIONAL on the present-epoch normalization UN (matter dilutes, vacuum constant — UN is a Σ_0 / charge–flux selection, not a local identity; §16.6). Within-matter 6:32 is single-trace-protected (DERIVED); cross-carrier 38:83 is gate XB-6b (OPEN → ZS-A29). The absolute normalization is reassigned to B3-B.

So the original B3 — the absolute late-time scale of ZS-A27 — is **not closed**. What this paper does is **reframe** (the ~120-order hierarchy *factorizes*), **ground** (the rank coefficient 83/121 via ZS-F2→U4→A19→A20), **localize** (the residue to named OPEN items), and **conditionally derive the identification bridge B3-D** (§16): its field-theory half is IMPORTED-PROVEN, and its identification half is an explicit DERIVED-CONDITIONAL implication on {VS, TFS, UN, PT}, with a parent action written (§16.4) and the gates named. v2.0 adds two refinements the reviews required: it separates the *rank* fraction **q**$\Lambda = 83/121$ (closed, mod PT) from the *energy* fraction **Ω**$\Lambda,0 = 83/121$ (conditional), so the parameter-free status of the flagship number is honestly tied to the present-epoch normalization; and it shows UN cannot be a local identity (matter dilutes, vacuum is constant), re-casting it as the present-epoch charge–flux selection — the cosmological coincidence problem — which is handed to ZS-A29 (§17). The remaining load-bearing OPEN items are the four B3-D antecedents — chief among them XB-6b — and, terminally, the absolute scale in B3-B. The correction history is honest in direction (each version demotes the previous version's over-statement while preserving the genuine imports), which is itself the argument, recorded in §17, for concluding A28 here and moving the crux to a dedicated follow-up plus external review.

Note on Part I's independence. For an external reader, Part I (§§2–9) is summarized here with its detailed derivations incorporated by reference to v1.7. For formal publication this should be resolved one of two ways: restore Part I's essential theorems and proofs in full, or publish Part I as a standalone companion No-Go paper (*Why microscopic vacuum-energy scale-generation fails*). v2.0 flags this as a publication-format obligation, not a physics gap.

§16. The Vacuum-Identification bridge B3-D: a projector-valued top-form, in two halves (field-theory half IMPORTED-PROVEN; identification half DERIVED-CONDITIONAL)

This section treats B3-D in two halves. The **field-theory half** — *that the rank-83 trace complement, when it carries any stress at all, carries it as a constant $w = -1$ vacuum stress tensor* — is established by importing the established top-form theorems (§16.2). The **identification half** — *that the Z-Spin vacuum is precisely this top-form, with the rank fraction promoted to the present energy fraction* — is a conditional implication resting on an explicit set of named antecedents (§16.3–§16.6). v1.9 corrected the v1.8 claim that the single-trace structure protected the cross-carrier ratio and wrote the unified action that isolates the residual free coupling (§16.4); v2.0 adds the structural-dilemma argument that the correction was *forced* (§16.3) and shows that the residual Unified-Normalization cannot be a local identity, re-casting it as a present-epoch selection (§16.6). The result is **not** a new derivation of 83, and **not** a determination of the absolute scale of ρΛ. The full deep-exploration record — Passes A (v1.8), B (v1.9), and C (v2.0) — is Appendix D.

§16.1 The reframe: do not re-explain 83; promote PΛ to a top-form support space

B3-D as stated in v1.7 reads: *that the trace complement $q\Lambda = 1 - qb - qc$ has $\tau(q\Lambda) = 83/121$ is closed arithmetic; that this complement is the physical vacuum stress tensor is the unproven bridge.* The decisive move is to stop treating B3-D as "explain the number 83 more" and instead treat it as a **carrier-identification** problem: *which spacetime field lives on the 83-dimensional complement, and what stress tensor does it produce?* Once posed this way, the closure follows from a theorem the corpus already imports.

Define, in the 121-dimensional channel algebra of ZS-A19/A20 (the face factor **A**face = End($V_{11} \otimes V_{11}$) ≅ $\mathbf{M}_{121}(\mathbb{C})$, equivalently End($\mathbb{C}^{121}$) ≅ $\mathbf{M}_{121}(\mathbb{C})$, built on the (3,2,6)$^{\otimes}$(3,2,6) register), the matter and vacuum projectors

Pb = the rank-6 baryon (XZ / cube) projector (ZS-A19 §10.2, ZS-F5 B3.1), **P**c = the rank-32 CDM (truncated-icosahedron) projector (ZS-A19 Thm A19.2, PROVEN), $\mathbf{P}\Lambda = \mathbf{I}_{121} - \mathbf{P}b - \mathbf{P}c$, with $\mathbf{P}b\mathbf{P}c = 0$. (A28.51)

Lemma 16.1 (Complement projector, PROVEN). With $\mathbf{P}b\mathbf{P}c = 0$ and ranks 6, 32, the complement **P**Λ is an orthogonal projector with $\mathbf{P}\Lambda^2 = \mathbf{P}\Lambda$, $\mathbf{P}\Lambda\mathbf{P}b = \mathbf{P}\Lambda\mathbf{P}c = 0$, and rank **P**Λ = 121 − 6 − 32 = 83, so $\tau_{121}(\mathbf{P}\Lambda) = 83/121$. *(Verified symbolically with explicit rank-6/32 projectors; verify_zs_a28_v1_8.py block D, checks D1–D6.)*

§16.2 The vacuum carrier is a top-form, and it forces w = −1 (IMPORTED-PROVEN)

The matter sectors **P**b, **P**c carry the *timelike* Brown–Kuchař densitized current of ZS-A20 ($\mathbf{J}\mu = j\mu(\mathbf{P}b + \mathbf{P}c + \cdots)$ with jμ timelike), which gives pressureless dust $\mathbf{T}\mu\nu = \rho m\, u\mu\, u\nu$, $w = 0$ (ZS-A20 §9, DERIVED on C1-ID-global). The complement **P**Λ is, by hypothesis (the Vacuum-Support Lemma, §16.5), *not* in the support of that timelike current: on **P**Λ there is no particle-number current and hence no preferred timelike direction. A perfect-fluid stress tensor with *no* preferred rest frame must have ρ + p = 0, i.e. $w = -1$ and $\mathbf{T}\mu\nu \propto \mathbf{g}\mu\nu$ — but this symmetry remark *alone* does not exclude non-perfect-fluid effective stresses, so we do **not** rest the result on a uniqueness argument. Instead we *select* a definite field on the **P**Λ corner — a **top-form** — and read its stress tensor off the action by metric variation. The top-form is the natural choice because it is the field whose four-form strength has, in four dimensions, no local propagating polarization and a *forced* on-shell constancy; a constant scalar potential V_0 also yields $w = -1$ but must have its constancy imposed by hand, whereas the top-form derives it (this distinction is the Top-Form-Selection question, §16.5).

Concretely, assign to the **P**Λ corner a 3-form gauge potential and its 4-form field strength,

$\mathbf{A}_3 = A_3\,\mathbf{P}\Lambda$, $\mathbf{F}_4 = \mathrm{d}\mathbf{A}_3$, $F\mu\nu\rho\sigma = f\,\varepsilon\mu\nu\rho\sigma$ on-shell.   (A28.52)

We then import three established theorems about a 4-form field strength in four dimensions [23–26]:

(T1) **No local degrees of freedom.** In 4D a 3-form gauge potential has no propagating polarizations; F_4 is non-dynamical. (T2) **Constancy on-shell (Henneaux–Teitelboim).** The Euler–Lagrange equation $\partial\mu(\sqrt{-g}\, F\mu\nu\rho\sigma) = 0$ forces f = const; equivalently the conjugate "cosmological constant" is a constant of the motion, $\partial\mu\rho\Lambda = 0$. This is the Henneaux–Teitelboim canonical structure [25]. (T3) **Vacuum stress tensor.** Varying the metric in the Maxwell-type 4-form action gives **T**μν**^(F) = −ρ**Λ **g**μν with $\rho\Lambda \propto f^2$, so $p\Lambda = -\rho\Lambda$ and **wΛ = −1** exactly.

Theorem 16.2 (Projector-valued top-form vacuum, field-theory half DERIVED / IMPORTED-PROVEN). Let the 121-channel face factor $\mathbf{M}_{121}$ carry the matter projectors (A28.51) and a 3-form potential $\mathbf{A}_3$ valued in the **P**Λ corner (A28.52). Then, by (T1)–(T3),

$\partial\mu\rho\Lambda = 0$,   **T**μν**^Λ = −ρ**Λ **g**μν, $w\Lambda = -1$, $p\Lambda = -\rho\Lambda$,   (A28.53)

and these three statements — constancy, the form of the stress tensor, and the equation of state — are *forced*, not assumed. The corpus's own $w = -1$ de Sitter attractor (ZS-U4 §9, DERIVED) is thereby *carried by the 83-block*, rather than posited as an independent input. *(Checks D7–D11; the w = −1 result is convention-independent — the overall prefactor that sets the value of ρΛ is exactly the central-shift offset discussed in §16.4.)*

This is the **field-theory half**, and it is solid: the v1.7 OPEN item "the complement *is* the physical vacuum stress tensor" is replaced by the IMPORTED-PROVEN conditional statement "the complement, *when it is the support of a top-form*, *necessarily* produces a constant $w = -1$ vacuum stress tensor." What remains conditional is the *antecedent* — that the complement is indeed the support of such a top-form (the Top-Form-Selection Lemma, §16.5) — not the implication.

§16.3 What the single trace protects (6:32) and what it does not (38:83): the cross-carrier correction

The obvious objection is that the coefficient 83/121 can be rescaled into a field normalization, **T**μν = $-\lambda_0(83/121)$**g**μν with λ_0 free, so that 83/121 is mere bookkeeping. We must answer this carefully, because v1.8 answered it *too strongly*. The corpus tool is **ZS-A20 §9.3**: in the **single-trace** matrix current

SZ = −μZ ∫d^4x Tr √(−gμν **J**μ **J**ν),   ρi = μZ *n* Tr **P**i,   (A28.54)

an independent invariant μb Tr **P**b + μc Tr **P**c with *separate* μb, μc *is forbidden* — there is one coefficient μZ, not one per sector. This fixes the ratio of any two sectors **that live inside the same current J**μ. Baryon and CDM do: both **P**b and **P**c are carried by the one Brown–Kuchař dust current, so

$\rho_b : \rho_c = \mathrm{Tr}\, \mathbf{P}_b : \mathrm{Tr}\, \mathbf{P}_c = \mathbf{6 : 32}$, (A28.55)

invariant under the only rescaling available, $\mu_Z \to \alpha\, \mu_Z$ (which multiplies *both* equally). This within-matter protection is **DERIVED** (verify block C, check D15) and is exactly what ZS-A20 proves.

The v1.8 overclaim, corrected. v1.8 extended (A28.55) to the *vacuum* and asserted that 6 : 32 : 83, hence the matter-to-vacuum ratio 38 : 83, is single-trace-protected. **This does not follow**, because the vacuum is *not* carried by $\mathbf{J}_\mu$. By Theorem 16.2 the vacuum is a *separate* top-form with its own coupling, contributing

$\rho_\Lambda = (Z_F / 2)\, f^2\, \mathrm{Tr}\, \mathbf{P}_\Lambda = 83 \cdot (Z_F / 2)\, f^2$, (A28.56)

with normalization (Z_F, f) *independent* of μ_Z. Hence

$\rho_{matter} / \rho_\Lambda = (38\, \mu_Z\, n) / (83 \cdot Z_F\, f^2 / 2) = (38 / 83) \cdot (2\, \mu_Z\, n / Z_F\, f^2)$, (A28.57)

and the bracket $(2\, \mu_Z\, n)/(Z_F\, f^2)$ is a **free cross-carrier ratio** — unaffected by $\mu_Z \to \alpha\, \mu_Z$, and changed by any independent rescaling $f \to \beta\, f$ of the vacuum coupling (verify block C, check D16: the residual factor is exactly $2\mu_Z n/(Z_F f^2)$). The single trace removes μ_b/μ_c *within a carrier*; it says nothing about the relative normalization of *two different carriers* (a Brown current and a top-form). **Therefore the 38:83 ratio is OPEN, not DERIVED.** It is promotable to **DERIVED-CONDITIONAL** only under the explicit

Condition UN (Unified-Normalization). $Z_F\, f^2 = 2\, \mu_Z\, n$ (equivalently, the top-form and the Brown current descend from a *single* parent action with one overall coupling).

Under UN, (A28.57) collapses to $\rho_{matter} / \rho_\Lambda = 38/83$ and the energy fraction $\mathbf{\Omega}_{\Lambda,0} = 83/121$ returns (verify check C4). UN is the cross-carrier analogue of A20's single trace, but it is a *condition to be derived*, not a theorem in hand; its derivation is gate XB-6b, and §16.5 shows it cannot be a *local* identity.

Why the demotion is forced, not optional (the structural dilemma). The correction is not a matter of taste: v1.8's two claims are *mutually exclusive*. To get $\mathbf{w = -1}$ the vacuum must be a top-form, i.e. a carrier *distinct* from the Brown current — but that distinctness is *exactly* what frees the 38:83 ratio (a second carrier brings a second coupling, A28.57). The only way to keep 38:83 single-trace-protected would be to put the vacuum *inside* the one Brown current; then all of 6:32:83 share one coefficient — but they also share one *dust* equation of state $w = 0$, and the top-form $w = -1$ argument is lost (verify block G). One cannot have both. v1.9 chose the correct branch — keep the top-form (and $w = -1$), demote the ratio — and v2.0 records that this was forced by the carrier structure, resolving an internal contradiction in v1.8 rather than merely tightening wording.

§16.4 The unified action (written explicitly) and what it does and does not fix

v1.8 declared $\mathbf{A}_3 = A_3 \mathbf{P}_\Lambda$ and imported the top-form theorems but never wrote the complete action governing B3-D. We do so now. The minimal action that contains both carriers is

$\mathbf{S} = \mathbf{S}\mathrm{grav}[g] + \mathbf{S}\mathrm{Brown}[\mathbf{J}\mu, \mathbf{P}\mathrm{b}, \mathbf{P}\mathrm{c}] + \mathbf{S}\mathrm{F}[\mathbf{F}_4, \mathbf{P}\Lambda] + \mathbf{S}\mathrm{constraint}[\mathbf{P}\mathrm{b}, \mathbf{P}\mathrm{c}, \mathbf{P}\Lambda]$, (A28.58)

with the Brown dust term **S**Brown of (A28.54), the projector-valued Maxwell top-form term

$\mathbf{S}\mathrm{F} = -(\alpha / 2\cdot 4!) \int d^4x \sqrt{-g}\, \tau_{121}(\mathbf{P}\Lambda\, \mathbf{F}\mu\nu\rho\sigma\, \mathbf{F}\mu\nu\rho\sigma)$, (A28.59)

and **S**constraint enforcing $\mathbf{P}\mathrm{b}^2 = \mathbf{P}\mathrm{b}$, $\mathbf{P}\mathrm{c}^2 = \mathbf{P}\mathrm{c}$, $\mathbf{P}\mathrm{b}\mathbf{P}\mathrm{c} = 0$, $\mathbf{P}\Lambda = \mathbf{I} - \mathbf{P}\mathrm{b} - \mathbf{P}\mathrm{c}$ (Lagrange multipliers in the 121-channel). The required variations are $\delta\mathbf{A}_3$ (Euler–Lagrange for the 3-form $\Rightarrow \partial\mu(\sqrt{-g}\, \mathbf{F}\mu\nu\rho\sigma) = 0$, constancy), $\delta g\mu\nu$ ($\Rightarrow \mathbf{T}\mu\nu = \mathbf{T}\mu\nu\text{\^{}Brown} + \mathbf{T}\mu\nu\text{\^{}F} = \rho m\, u\mu u\nu - \rho\Lambda\, g\mu\nu$), and $\delta\mathbf{P}$ subject to the projector constraints (fixes the *supports*, i.e. which corner each field occupies).

Writing (A28.58) makes the situation exact and unflattering in the right way. The variation δg confirms the field-theory half (the top-form piece is $-\rho\Lambda g\mu\nu$, $w = -1$). But (A28.58) contains **two independent couplings**, μZ in **S**Brown and α (with *f*) in **S**F, and *nothing in the action relates them*: the cross-carrier ratio (A28.57) is genuinely free. The single-trace structure lives *inside* **S**Brown and does not reach across to **S**F. So the explicit action does not *close* the cross-carrier ratio — it *isolates* it as the single coupling that Condition UN (gate XB-6b) must fix. This is the honest content of the unified action: it converts a vague "the ratio is protected" into a precise "here is the one free coupling, and here is the condition that would fix it."

What the action does settle cleanly is the **absolute** scale: the common normalization of (A28.58) — equivalently the value of ρΛ itself — is the additive vacuum offset against which, by the **ZS-A26 Central-Shift No-Go**, *every relative selector is invariant* (it factors through the central-shift quotient A/R_center, $\Gamma\mathrm{eff} \to \Gamma\mathrm{eff} + c_0\int\sqrt{g}$). By ZS-A19.NG2 the projector ranks cannot determine it. **We therefore reassign the absolute scale from B3-D to B3-B**, where the A26 no-go locates it — this part of v1.8 was correct and is retained.

§16.5 The residual is a set, not a single debt: the antecedents of B3-D

v1.8 stated that B3-D's residual is the *single* Vacuum-Support Lemma. That is also corrected: the identification half rests on a **set** of named antecedents, no one of which implies the others.

Condition VS (Vacuum-Support Lemma). The timelike Brown–Kuchař matter current is supported only on the matter block: $s(\mathbf{J}\mathrm{matter}) = \mathbf{P}\mathrm{b} + \mathbf{P}\mathrm{c}$, so that $\mathbf{P}\Lambda = \ker \mathbf{J}\mathrm{matter}$ is current-null.

Condition TFS (Top-Form-Selection Lemma). The field occupying the current-null corner *is* a top-form: supp $\mathbf{F}_4 = \mathbf{P}\Lambda$. *VS does not imply TFS*: "no matter current lives here" does not entail "a top-form lives here" (a constant scalar potential, or simply nothing, are a priori compatible with current-nullity). TFS is the carrier-selection step, **HYPOTHESIS-strong** (motivated by no-local-d.o.f. + forced constancy, not forced), gate XB-6c.

Condition UN (Unified-Normalization, present-epoch form). The matter and vacuum normalizations satisfy ZF $f^2 = 2$ μZ n_0 *on the present-epoch hypersurface* Σ_0 — **not** as a local identity for all times (§16.6 shows the local identity is impossible). This is the condition that promotes the rank fraction to the energy fraction **Ω**Λ,0 = 83/121 today. **OPEN**, gate XB-6b (renamed *Present-Epoch Boundary Charge–Flux Matching*).

Condition PT (Physical-Trace$_{121}$). The physical-quotient denominator is 121, not 120 (the Case A vs Case B question of §12): it shifts the *value* 83/121 ↔ 82/120, **OPEN**, but does not touch the $w = -1$ structure.

On VS itself, the v1.8 status "DERIVED-CONDITIONAL on ZS-U4" is **too strong**: U4 *adopts* the (6, 32, 83)/121 budget as a locked late-time input, but adopting a budget does not *prove* the action-level statement s(**J**matter) = **P**b + **P**c (that the *only* timelike-current carriers are baryon and CDM). The honest status is **DERIVED-CONDITIONAL on an explicit Exhaustive-Matter-Content axiom** (that no further pressureless carrier exists at late times), with the A20 §9.1–9.2 promotion path (derive the support from a first-order parent action; the local statement Jμ = ∂νΣμν reproduced the Brown identities for ΣZ). This is gate XB-6a. The global flux quantization fixing the integer ranks as cohomology classes is the *same* C1-ID-global gate ZS-A20 §9.5 already flags (∗**J** = d**B** + Σ qA ωA, ωA ∈ H^3(MZ, ℤ)); B3-D inherits it without adding a *new* debt.

§16.6 Unified-Normalization is not a local identity: the present-epoch / charge–flux form (v2.0)

v1.9 stated Condition UN as the *local* identity ZF $f^2 = 2$ μZ n. v2.0 corrects this: **the local identity cannot hold.** The Brown–Kuchař dust number density dilutes with expansion, $n(t) \propto a(t)^{-3}$ (ZS-A20 §9; verify D1), so the matter density redshifts, ρm(a) = 38 μZ $n_0\, a^{-3}$. The top-form field strength is a spacetime constant on-shell, $\partial_\mu f = 0$ (Theorem 16.2; verify D2), so the vacuum density ρΛ = 83 (ZF/2)f^2 is a-independent. Hence the proposed identity has a constant left-hand side and a right-hand side $\propto a^{-3}$ (verify D3); it can be satisfied on **at most one hypersurface** $a = a_0$ (verify D4). Equivalently, the energy ratio

ρmatter(a) / ρΛ = (38/83) · (2 μZ n_0 / ZF f^2) · a^{-3} (A28.61)

equals the rank ratio 38/83 at a *single epoch* (verify D5), and the energy fraction **Ω**Λ(a) = ρΛ /(ρΛ + ρm(a)) equals the rank fraction **q**Λ = 83/121 only there (verify E1–E3). **The rank fraction is a time-independent fact about the projector; the energy fraction is a time-dependent cosmological observable; they meet at one moment.** This is precisely the cosmological coincidence ("why now?") problem, here stated in Z-Spin terms: *why is the present epoch the one at which the energy fraction coincides with the rank fraction?*

Consequently UN must be re-read as one of three *non-local* statements, all equivalent to selecting the present-epoch hypersurface Σ_0:

(i) **present-epoch boundary condition** — ZF $f^2 = 2$ μZ n_0 on Σ_0; (ii) **global charge–flux relation** — ZF $f^2 \propto$ μZ Nm / $V_3(\Sigma_0)$, tying the top-form flux to the integrated matter charge and the boundary three-volume; (iii) **four-volume-clock condition** — the Henneaux–Teitelboim four-volume time (or a BV–BFV boundary state) whose eigenvalue selects Σ_0.

These reframe gate XB-6b from a coupling match into the **Present-Epoch Boundary Charge–Flux Matching** problem, which is the genuine physical bridge between **B3-C** (the present-day relation, which already takes H_0 — equivalently the Σ_0 clock reading — as input) and **B3-D** (the carrier identification). Because B3-C is itself DERIVED-CONDITIONAL on H_0, this is consistent: the present-epoch selection is the same datum as the B3-C "today" input, not a new free parameter hidden inside B3-D. Whether the four-volume clock or boundary state *derives* Σ_0 (rather than merely labelling it with the observed value) is OPEN, decidable, and the central subject handed to **ZS-A29** (§17).

§16.7 Status summary for B3-D (v2.0)

Item	Status after §16 (v2.0)	Basis
Complement projector: rank PΛ = 83	**PROVEN**	Lemma 16.1; verify A1–A8
Rank fraction qΛ = τ_{121}(PΛ) = 83/121	**PROVEN, DERIVED-COND on PT** (denominator 121 vs 120)	§12; verify A8, F1–F3
Top-form ⟹ ∂μρΛ = 0, Tμν = −ρΛgμν, w = −1 (given the Maxwell four-form action)	**IMPORTED-PROVEN**	[23–26]; Theorem 16.2; verify B1–B2
Within-matter ratio ρb : ρc = 6 : 32	**DERIVED**	ZS-A20 §9.3 single trace (same carrier); verify C1
Cross-carrier ratio 38 : 83 (rank → energy)	**OPEN** (DERIVED-COND under UN)	A28.57; free (ZF f²)/(2μZn); verify C2–C4; gate XB-6b
Energy fraction ΩΛ,0 = 83/121 (today)	**DERIVED-CONDITIONAL on {VS, TFS, UN, PT, Frame}**	§12, §16.6; the flagship-prediction status
Top-form-Selection (supp F_4 = PΛ)	**HYPOTHESIS-strong**	motivated, not forced; gate XB-6c
Vacuum-Support (PΛ = ker Jmatter)	**DERIVED-COND on an explicit Exhaustive-Matter-Content axiom**	not "on U4"; gate XB-6a

Item	Status after §16 (v2.0)	Basis
Unified-Normalization UN — *local* identity	**RETRACTED (impossible)**; replaced by present-epoch / charge–flux form	§16.6; verify D1–D5
Unified-Normalization UN — present-epoch form	**OPEN** = the coincidence/selection problem	gate XB-6b → ZS-A29
Physical-Trace PT (121 vs 120)	**OPEN, decidable** (BRST/BV–BFV cohomology)	§12; both within ≈1% of data
Structural dilemma (top-form XOR single-trace 38:83)	**PROVEN** (mutually exclusive)	verify G1–G2
Radiative stability / sequestering	**PARTIAL**	inherited from A27 R0
Absolute scale (value of $\rho\Lambda = c_0$)	**OPEN → reassigned to B3-B**	ZS-A19.NG2, ZS-A26
B3-D field-theory half	**IMPORTED-PROVEN**	top-form theorems on PΛ
B3-D identification half	**DERIVED-CONDITIONAL** on {VS, TFS, UN, PT}	implication A28.60
B3-D as an unconditional closure	**PARTIAL**	gate XB-6 pending → ZS-A29

The defensible deliverable is the conditional implication, stated once and completely:

[**VS** + **TFS** + **UN**(Σ_0) + **PT**$_{121}$] $\Rightarrow$ **T**μν**^Λ = −ρ**Λ **g**μν, *w* = −1, with **Ω**Λ,0 = 83/121.
(A28.60)

Net. v2.0 keeps the v1.9 corrections and sharpens two further points the reviews surfaced. First, the *rank* fraction **q**Λ = 83/121 (a projector fact, closed modulo the decidable PT denominator) is cleanly separated from the *energy* fraction **Ω**Λ,0 = 83/121 (a cosmological observable, DERIVED-CONDITIONAL); the parameter-free status of the flagship number therefore *rests on* the present-epoch normalization UN — stated openly. Second, UN is shown to be impossible as a local identity (matter dilutes, vacuum is constant) and is re-cast as a present-epoch / charge–flux selection — the cosmological coincidence problem — which is the true bridge between B3-C and B3-D. The field-theory half remains IMPORTED-PROVEN; B3-D as an unconditional closure remains PARTIAL; and the crux (the present-epoch 38:83 selection, the single-parent action that might derive VS/TFS/UN together) is handed to ZS-A29 with an explicit program.

§17. What A28 v2.0 concludes, and what passes to ZS-A29

The reviews that shaped v2.0 also reached a structural recommendation: A28 has, with the present consistency fixes, reached the level of closure attainable *inside this paper*, and the remaining crux is better treated as its own work than by further internal iteration of A28. v2.0 adopts that recommendation.

Concluded in A28 (the audit of the microscopic and boundary routes). Part I's negative result (every internal scale engine re-expresses one number N, A–Q-Only-forbidden) is settled CLOSED-NEGATIVE. Part II's reframe ($\rho_\Lambda^{1/4} = \sqrt{\bar{M}_P H_0}$; the hierarchy *factorizes* into the rank coefficient $\mathbf{q}_\Lambda$ plus the B3-B absolute scale) is the standing picture. The B3-D field-theory half (top-form $\Rightarrow w = -1$, constancy, $T = -\rho\mathbf{g}$) is IMPORTED-PROVEN. The rank fraction $\mathbf{q}_\Lambda = 83/121$ is closed modulo the decidable PT denominator. The absolute scale is terminally in B3-B (ZS-A26 No-Go). These do not need re-opening.

Passed to ZS-A29 — *A Single-Parent Measure/Form Action for Dust and Vacuum: Global Charge–Flux Matching, the Present-Epoch Selection Problem, and the 38:83 No-Go or Closure.* The single load-bearing question that decides the parameter-free status of the flagship energy fraction is: *given that dust dilutes ($\rho_m \propto a^{-3}$) while the top-form vacuum is constant, what selects the present epoch at which the energy fraction equals the rank fraction 83/121?* This is the present-epoch form of UN (§16.6, gate XB-6b). A29's program, in decreasing decidability:

(1) **Boundary Charge–Flux Matching (No-Go or closure).** Prove the local identity $Z_F f^2 = 2\,\mu_Z\, n(t)$ impossible (done here, §16.6), then test whether a Henneaux–Teitelboim four-volume clock or a BV–BFV boundary state *selects* Σ_0 from a global relation among the matter charge N_m, the top-form flux Q_F, and the boundary three-volume V_3 — and crucially whether that selection is a derivation or a tautology that re-inputs the observed value.

(2) **Single-parent measure/form action.** Test whether a non-Riemannian-measure action of the Guendelman–Nissimov–Pacheva type [27,28] — in which a conserved dust current and a dynamically generated cosmological term arise *together* from one covariant action via a hidden nonlinear Noether symmetry — can host the Z-Spin projectors $\mathbf{P}_m = \mathbf{P}_b + \mathbf{P}_c$ and $\mathbf{P}_\Lambda = \mathbf{I} - \mathbf{P}_m$ so as to derive **VS**, **TFS**, and **UN** jointly. Success would promote B3-D toward DERIVED; failure would yield a sharp No-Go (one action generating both dust and Λ still leaves their *present-epoch* ratio fixed only by integration constants).

(3) **Physical-Trace PT (121 vs 120).** The most finite gate: a BRST / BV–BFV [29] physical-state cohomology computation of whether the Z_2-odd removed mode is gauge-exact (denominator 120, $\mathbf{q}_\Lambda = 82/120$) or a physical background mode (denominator 121, $\mathbf{q}_\Lambda = 83/121$), combined with the A20/A21 seam cohomology.

(4) **Top-Form-Selection theorem or No-Go**, and (5) **radiative stability** once a parent action is fixed.

The honest meta-point the review record makes — that each A28 version has caught the previous version's overclaim, so internal iteration converges to *honesty* but not to *closure* — argues that the highest-value next steps are not a v2.1 but (a) the loop-resistant decidable computations above, executed where possible *outside* the author-plus-AI loop, and (b) external adversarial review of a narrow, jargon-reduced statement of the data channels ($\Delta N_{eff}^{BBN} = 0.160$, $r = 0.0089$, $\mathbf{\Omega}_{\Lambda,0} = 83/121$). The v2.0 audit is itself the evidence for (b): an independent review caught a real error two versions running.

Code availability and an honest verification ledger. The attached **verify_zs_a28_v2_0.py** is a *fail-closed* script (every check is a real Python assert; any failure raises and the process exits nonzero) covering the **27 load-bearing B3-D asserts**: block A (A1–A8, complement-projector algebra), block B (B1–B2, the top-form $w = -1$), block C (C1–C4, single-trace *scope* — within-matter 6:32 protected, cross-carrier 38:83 free, UN recovers it), block D (D1–D5, the v2.0 time-dependence result: matter dilutes $\propto a^{-3}$, the vacuum is constant, so UN cannot be a local identity), block E (E1–E3, the rank fraction $\mathbf{q}_\Lambda$ is a-independent while the energy fraction $\mathbf{\Omega}_\Lambda(a)$ is a-dependent), block F (F1–F3, 83/121 vs 82/120 vs observation), and block G (G1–G2, the structural dilemma). This script *is* attached and independently runnable, and supersedes the v1.9 core-audit script. **Scope of the verification claim.** This number is the *load-bearing audit*, not the full corpus regression suite: the ≈111 inherited v1.7/v1.9 checks (Part I retraction/symplectic/susceptibility; Part II geometric-mean/holographic budget; the v1.5 repair block; the v1.6 lineage and ΔN_{eff} correction) are maintained separately in the Z-Spin repository and are *not* reproduced in this manuscript. Earlier drafts stated a single "111/111" figure on the cover; v2.0 instead states the attached fail-closed audit count (27/27) and flags the full suite as external — a deliberate honesty fix, since a reader can run the 27 here but cannot, from this file alone, confirm the inherited 111. **Methodological note.** The right epistemic frame for a *zero-parameter* theory is two distinct judgements: the mathematics is decidable (right or wrong by computation, subject to *independent adversarial error-checking* because author and AI can share blind spots), and the parameter-free predictions are *falsified or confirmed* by data, not fitted. A passing suite confirms only internal consistency. **On the loop — and two worked examples of it.** v1.9 caught the v1.8 single-trace overclaim (the same shape of over-reach the corpus earlier caught in ZS-A19 v1.0, "32 faces → dark matter"); v2.0 in turn caught two residual over-statements in v1.9 — the §12 *energy*-fraction claim stated unconditionally, and UN stated as a *local* identity — both flagged by independent review and confirmed by the symbolic checks above (blocks D, E). The pattern is explicit: each version's "closure" is partly walked back by the next version's audit, so **internal iteration converges to honesty, not to closure**. The implication, recorded in §17, is that the highest-value next steps are not a v2.1 but the loop-resistant decidable computations of ZS-A29 (executed where possible outside the author-plus-AI loop) and external adversarial review of a narrow, jargon-reduced statement of the data channels ($\Delta N_{eff}^{BBN} = 0.160$, $r = 0.0089$, $\mathbf{\Omega}_{\Lambda,0} = 83/121$). This work used Anthropic Claude; the author assumes full responsibility, including for the errors caught only on later review.

Appendix A. The symplectic-selection computation (Theorem A28.7) — preserved from v1.7.

In the basis $h = I$, $C = J_0K$, a general anti-Hermitian generator is $X = [[ia, b],[-\bar{b}, ic]]$; commutation with the antilinear C reduces to $a = -c$, leaving the traceless anti-Hermitian matrices ≅ $su(2)$; the exponential is SU(2) = USp(2). The construction is the statement that $\mathbb{C}^2$ with a quaternionic structure is $\mathbb{H}$, and U(1, $\mathbb{H}$) = Sp(1) = SU(2).

Appendix B. External-anchor map and cross-version safety — preserved from v1.7, extended for v2.0.

A28 v2.0 *uses* (never modifies) ZS-M27 ($\mathbb{C}^2$ odd cohomology), ZS-Q14 (spin-½, $\chi Z = -1$, Kramers *C*), ZS-S15 ($JZ = \gamma^5$), ZS-F9 ($\rho Z = 0$), ZS-F23 (the $\mathbb{Z}_2$-seam ln 2), ZS-A19/A20 (the rank-32/6 projectors, the single-trace matrix current, A19.NG1/NG2, C1-ID-global), ZS-A26 (the central-shift No-Go), ZS-U4 (the locked (6,32,83)/121 budget; w = −1 attractor), and ZS-A24–A27 (the B3 chain). *z** (ZS-M1) is untouched. **Cross-version conflict check (v2.0).** No corpus paper is contradicted, and the v2.0 fixes *strengthen* fidelity. A26's central-shift No-Go is respected; A27's A–Q-only No-Go is respected; A19.NG1/NG2 are respected/used; A20's single-trace scope is now applied *only within one carrier* (the v1.9 fix), and the v2.0 §12 *rank-vs-energy* split makes this sharper still — the corpus *rank* budget (6,32,83)/121 is a face-counting fact, and the *energy* fractions are explicitly conditional on the present-epoch normalization, so no corpus statement about ΩΛ is now over-read. Crucially, the v2.0 observation that $\rho m \propto a^{-3}$ while $\rho\Lambda$ = const is *exactly* ZS-A20's own dust scaling and ZS-U4's own w = −1 attractor — the present-epoch reframe of UN is therefore *consistent with*, not a revision of, A20/U4; it merely makes explicit that the rank/energy coincidence is an epoch-dependent statement. A20's C1-ID-global is the shared topological gate. No downstream value (η_B, r, ΔN_eff, z*) changes.

Result / gate	External PROVEN anchor	Role
SU(2)_Z (B3-4, §5)	compact symplectic group Sp(1)=SU(2); Kramers $T^2=-1$	forces the odd gauge group from spin-½
susceptibility (B3-2/5, §4)	positivity of pure-gauge topological susceptibility (lattice)	rank ≥ 1; sets residual coefficient
R0 (B3-1, §3)	Kaloper–Padilla / Omnia-Sequestra [3,4]; Klinkhamer–Volovik [5–7]	removes constants; EW-kick residual
B3-D vacuum bridge (§16)	**top-form / 4-form theorems [23–26]: no local d.o.f., constant on-shell, T = −ρg, w = −1**	**forces the vacuum structure (field-theory half) on the rank-83 complement PΛ; the identification half is conditional on {VS, TFS, UN, PT}**

Appendix C. The gravitational Pontryagin channel (demoted in v1.3) — preserved from v1.7.

Appendix D. Deep-exploration record for the B3-D closure: Passes A (v1.8), B (v1.9), C (v2.0) (5-step protocol)

This appendix records two passes. **Pass A (v1.8)** is the original brainstorm → MECE → issue-tree → convergence → scoring run that produced §16. **Pass B (v1.9)** is the *audit* of Pass A against two independent adversarial reviews — the run that produced the corrections in this version. Both are kept, because the audit only makes sense against the thing it audits, and because the correction is itself an instance of the protocol's self-reference discipline.

Pass A (v1.8) — origin, retained as record

Step 0 — Long list (7 candidate closure routes). (0.1) Insert a bare Λ term $S\_Λ = −∫\sqrt{−g}\, ρΛ$. (0.2) Identify a scalar-potential constant V_0 with 83/121. (0.3) **Henneaux–Teitelboim top-form** (3-form A_3, $F_4 = dA_3$). (0.4) Maxwell-type 4-form energy. (0.5) Klinkhamer–Volovik q-theory vacuum variable $q = *F_4$. (0.6) Vacuum-energy sequestering (radiative protection). (0.7) de Sitter entropy / holographic boundary law alone. **Dropped:** 0.1 (circular — assumes B3-D); 0.2 (blind to the additive constant — the A26 central-shift No-Go) — *note, flagged again in Pass B as the live alternative to TFS*; 0.7 (a *size* relation, not a *carrier* identification). **Retained:** projector algebra (0.3-support), Henneaux–Teitelboim top-form (0.3, main), Maxwell 4-form (0.4, cross-check), q-theory (0.5, stress-tensor cross-check), sequestering (0.6, radiative layer).

Step 1 — Issue list (4 MECE issues). (I) the 121-channel face factor and its trace; (II) why **P**Λ is "vacuum" (current-null maximality); (III) the covariant top-form action connecting **P**Λ to w = −1; (IV) normalization / radiative protection. **Dropped from B3-D:** absolute H_0 generation (B3-B), observational fit (downstream), the A27 Z_2-odd operator.

Step 2 — Issue tree. B3-D = [$qΛ → T_{μν}Λ = −ρΛ g_{μν}$], branches I → II → III → IV.

Pass B (v1.9) — the audit

Two external reviews converged on one defect. Review 1 (technical, detailed) identified the §16.3 single-trace claim as a *load-bearing error*, not a wording slip, and listed five further corrections (unified action unwritten; "single debt" wrong; VS status too high; "unique stress tensor" overclaim; record/format issues). Review 2 (lenient) independently flagged the same single-trace dependence as a corpus-conditional caveat and stressed the loop risk (the framing is AI-assisted). The audit question: **which Pass-A node statuses survive a decidable re-check?**

Step 3′ — Re-traversal with corrected epistemic status (changes from Pass A in **bold**):

- I-1 (M_{121} normalized trace τ = rank/121): **PROVEN, conditional on the *physical* M_{121} trace** (Review 1 §8 — the 121-vs-120 denominator question is upstream of this). I-2 (the 6/32/83 ranks): PROVEN arithmetic, DERIVED-CONDITIONAL on the face-factor identification. Typo fix: $\mathrm{End}(V_{11} \otimes V_{11}) \cong M_{121}$ (not M_{111}).
- II-1 (PΛ = I − Pb − Pc unique complement): PROVEN (Lemma 16.1).
- II-2 (Vacuum-Support, PΛ = ker Jmatter): **DERIVED-CONDITIONAL on an explicit Exhaustive-Matter-Content axiom** — *corrected from "on ZS-U4."* Adopting U4's budget does not prove action-level support-exhaustion (Review 1 §5).
- **II-2′ (Top-Form-Selection, supp F_4 = PΛ): NEW NODE, HYPOTHESIS-strong.** Split out from II-2 because *VS does not imply TFS* (Review 1 §4): current-nullity does not entail that a top-form (rather than a scalar V_0, or nothing) lives on the corner.
- III-1..4 (top-form variation: dλ = 0, T = −ρg, w = −1): **IMPORTED-PROVEN, given the Maxwell four-form action** — unchanged in content, but Theorem 16.2 is re-scoped as *conditional on the chosen action*, and the "unique stress tensor by symmetry alone" gloss is removed (Review 1 §6, Review 2 caveat 1).
- **IV-1 (ratio not reabsorbable): SPLIT.** IV-1a *within-matter* ρb:ρc = 6:32 — **DERIVED** (single-trace, same carrier; verify D15). IV-1b *cross-carrier* ρmatter:ρΛ = 38:83 — **OPEN** (free factor $(ZF f^2)/(2\mu Zn)$; verify D16), promotable to **DERIVED-CONDITIONAL under Condition UN** (verify D17). *This is the correction of the Pass-A "IV-1: DERIVED."*
- IV-2 (absolute scale): OPEN → B3-B (A19.NG2, A26) — unchanged, correct.
- IV-3 (radiative stability): PARTIAL — unchanged.

Step 4′ — Convergence of the audit. Pass A reported residual = {VS, denominator} (2 items) and claimed the ratio closed. The audit re-traversal *grows* the residual to the explicit set {VS, **TFS**, **UN**, PT-denominator} = 4, by (i) demoting IV-1b from DERIVED to OPEN and (ii) splitting TFS out of VS. First re-pass N_changed = 3 (IV-1b status; the VS→{VS,TFS} split; VS condition relabel). Does writing the unified action (§16.4) spawn a *fifth* independent condition? No: the action contains exactly two couplings (μZ, and α/f), and their unfixed ratio *is* UN — already counted; radiative stability and flux quantization were inherited in Pass A, not new. Second re-pass N_changed = 0 → **CONVERGED** to the 4-condition set. The MECE decomposition is revalidated at higher resolution (Pass A's single "protection" node was not atomic; it split into within- and cross-carrier).

Step 5′ — Scoring: a correction, not a promotion. The audit is convergent, corpus-non-conflicting (Appendix B; the correction *tightens* respect for A20's single-trace scope rather than conflicting with it), and **anti-numerology-positive in an unusual direction**: it *removes* a claimed protection (38:83), making the theory more conservative, and introduces no new tunable number. Per the protocol, this is not a HYPOTHESIS→DERIVED promotion; it is a DERIVED→OPEN *demotion* of one sub-claim with simultaneous confirmation (IMPORTED-PROVEN) of the field-theory half. Resulting status: **field-theory half IMPORTED-PROVEN; identification half DERIVED-CONDITIONAL on {VS, TFS, UN, PT}; B3-D as an unconditional closure PARTIAL.** Value scores, re-rated: corpus-internal integration 9.0/10 (the explicit unified action improves auditability); B3-D closure realism **6.5/10** (down from Pass A's 8.5 — the cross-carrier gap is now visible); mathematical clarity 9.0/10; external novelty

6.0/10; zero-parameter completeness **4.5/10** (down — UN is a newly explicit free coupling, not just the absolute scale).

The decidable core (verify_zs_a28_v1_9.py, block C — the audit that forced the correction).

```
rho_b = muZ*n*6 ; rho_c = muZ*n*32          # baryon, CDM: ONE Brown current

rho_m = rho_b + rho_c                       # = 38 muZ n

rho_L = (Z_F/2)*f**2*83                     # vacuum: SEPARATE top-form

# D15 within-matter, single-trace works:

assert simplify(rho_b/rho_c) == simplify((alpha*rho_b)/(alpha*rho_c))   # 6:32 invariant -> DERIVED

# D16 cross-carrier, single-trace does NOT reach:

assert simplify(rho_m/rho_L) == 76*muZ*n/(83*Z_F*f**2)                # free factor (Z_F f^2)/(2 muZ n)

assert simplify(rho_m/((Z_F/2)*(beta*f)**2*83)) != simplify(rho_m/rho_L) # changes under f->beta f -> OPEN

# D17 under imposed Unified-Normalization:

assert simplify(rho_m / rho_L.subs(Z_F, 2*muZ*n/f**2)) == Rational(38,83) # returns ONLY as a CONDITION
```

All three assertions pass, which is precisely why IV-1 had to be re-graded.

Self-reference and pattern audit (the loop, made concrete). Pass A guarded against the familiar-pattern risk *discrete rank + proven carrier action → cosmological fluid* by naming the Vacuum-Support Lemma rather than asserting "the leftover projector is therefore vacuum." That guard was correct but *incomplete*: it caught the existence-of-vacuum over-reach (now TFS) yet **missed a second, subtler over-reach** — assuming the single-trace protection that genuinely holds *within* the dust carrier also reaches *across* to a different carrier. This is the same shape of error the corpus earlier caught in ZS-A19 v1.0 ("32 faces, therefore dark matter"): a true statement about one structure silently extended one step too far. It was caught here only by two outside reviews plus the symbolic check above — exactly the loop pattern Review 2 named (the closure appeared the moment a reviewer pointed at the bridge). The honest reading: the deep-exploration protocol is good at *locating* residuals but, run inside the loop, is liable to *under-count* them by one; independent adversarial review and a literal symbolic re-check are the mechanisms that close the gap, and v1.9 is the worked example. The correct one-line statement is therefore: *the externally-proven top-form theorems close the field-theory half of B3-D; the Z-Spin-specific half reduces to the explicit four-condition implication (A28.60), of which the cross-carrier*

Unified-Normalization (XB-6b) is the most consequential and was the one v1.8 wrongly believed already closed; the absolute scale remains in B3-B.

Pass C (v2.0) — the audit of the audit

Two further independent reviews of v1.9 converged on a single structural point and several record fixes. The audit question for Pass C: **is v1.9's §12 (ΩΛ = 83/121) consistent with v1.9's §16 (38:83 OPEN), and what is the true content of UN given that matter dilutes but the vacuum is constant?**

Step 0 — Long list (7 candidates). (0.1) rank fraction qΛ vs energy fraction ΩΛ,0 — is §12 conflating them? (0.2) UN's time-dependence ($\rho m \propto a^{-3}$ vs ρΛ = const) — can UN be a local identity? (0.3) the structural dilemma (top-form-vacuum ⊕ single-trace-protection mutual exclusivity). (0.4) the PT denominator (121 vs 120) and the numeric value. (0.5) the flagship-prediction stakes (is ΩΛ still "zero-parameter"?). (0.6) the A28/A29 scope split. (0.7) record consistency (verification file, "dissolves," residual v1.8 tags). **Dropped from the epistemic tree:** (0.6) scope (a publication-strategy decision, handled as the §17 forward-pointer, not a status node) and (0.7) record hygiene (mechanical edits, no status impact).

Step 1 — Issue list (4 MECE issues, importance order). (A) UN's correct form — the time-dependence theorem and present-epoch reframe [from 0.2]. (B) qΛ-vs-ΩΛ,0 separation and the conditional status of the flagship number [0.1, 0.5]. (C) the structural dilemma — why the v1.9 demotion was forced [0.3]. (D) PT (the one decidable gate fixing the numeric value) [0.4].

Step 2 — Issue tree. C (dilemma) → A (UN form) → B (rank/energy status); D parallel. C is the root because it shows the carrier *must* be separate, which forces A's time-dependence; A determines B's conditionality; D is independent (it sets the value 83/121 vs 82/120 regardless of A/B/C).

Step 3 — Traversal with epistemic status.

- C (dilemma): **PROVEN** (verify G1–G2). The two v1.8 claims are mutually exclusive; v1.9's branch (keep top-form/w = −1, demote ratio) is the correct one.
- A (UN form): the *local* identity $Z_F f^2 = 2\mu_Z n(t)$ is **impossible** (LHS const, RHS $\propto a^{-3}$; verify D1–D5) ⇒ **RETRACTED** as a local identity. The correct replacement — a present-epoch / global charge–flux selection of Σ_0 (the coincidence problem) — is well-defined but its *derivation* is **OPEN** (whether a four-volume clock / boundary state selects Σ_0 rather than labelling it with the observed value). Hands to ZS-A29.
- B (rank vs energy): **qΛ** = τ_{121}(PΛ) = 83/121 is **PROVEN** (rank arithmetic), **DERIVED-CONDITIONAL on PT**. **ΩΛ,0** = ρΛ,0/ρtot,0 = 83/121 is **DERIVED-CONDITIONAL on {VS, TFS, UN, PT, Frame}** (verify E1–E3). The flagship prediction's parameter-free status is load-bearing on UN — a §12 status correction.
- D (PT): **OPEN, decidable** (BV–BFV/BRST cohomology). 83/121 = 0.6860 vs 82/120 = 0.6833; observed ≈ 0.6847, both within ≈ 1% (verify F1–F3); does not affect w = −1.

Step 4 — Convergence. Pass C reclassifies UN (local → present-epoch) and splits the §12 claim (rank vs energy); it does **not** spawn a fifth independent condition — the present-epoch selection *is* the content of UN, correctly stated, and it coincides with the B3-C "today"/H_0 input (Frame), not a new free parameter. N_changed first pass = 3 (UN reclassification; §12 split; dilemma made explicit); second pass = 0 (the present-epoch condition is stable — it is the single coincidence problem). **CONVERGED**, to a *cleaner* picture: the four antecedents are {VS, TFS, UN($\Sigma_0 \approx$ Frame), PT}, and the real crux is UN-present-epoch = ZS-A29's subject.

Step 5 — Scoring: again a correction + sharpening, not a promotion. Convergent, corpus-non-conflicting (Appendix B; the dilution reframe is A20/U4's own behaviour), anti-numerology-positive (it makes explicit that the *energy* fraction 83/121 is conditional, removing a possible over-claim; no new tunable number). Per the protocol, no promotion: q-fraction PROVEN(mod PT), energy-fraction DERIVED-CONDITIONAL, B3-D unconditional closure PARTIAL. Value re-rating: corpus-internal integration 9.2/10; B3-D closure realism 6.0/10 (the present-epoch gap is now explicit); mathematical clarity 9.2/10; zero-parameter completeness **4.0/10** (the flagship *energy* fraction is conditional on UN, a more honest score than v1.9's 4.5). Per the protocol's convergence rule, where the traversal converges but the central node (UN-present-epoch) remains OPEN, the honest outcome is "this cannot be closed with current corpus tools" — registered as such and handed to ZS-A29.

The decidable core (verify_zs_a28_v2_0.py, blocks D–G; 27 real fail-closed asserts).

```
a, n, muZ, ZF, f = symbols('a n mu_Z Z_F f', positive=True)

rho_m_a = muZ*(n*a**-3)*38      # dust dilutes ~ a^-3 (A20)

rho_L   = (ZF/2)*f**2*83        # vacuum constant on-shell (HT)

assert diff(rho_m_a, a) != 0 and diff(rho_L, a) == 0          # D1,D2 matter dilutes, vacuum const

assert diff(ZF*f**2, a) == 0 and diff(2*muZ*n*a**-3, a) != 0  # D3 local identity impossible for all a

q_L       = Rational(83,121)                              # rank fraction (a-independent)

Omega_L_a = simplify(rho_L/(rho_L+rho_m_a))                  # energy fraction (a-dependent)

assert diff(q_L,a) == 0 and diff(Omega_L_a,a) != 0           # E1,E2 rank != energy in general

assert len(solve(Eq(Omega_L_a, q_L), a)) >= 1                # E3 they coincide at one epoch (UN=Sigma0)

assert (p:=-Rational(1,2)*f**2)/(Rational(1,2)*f**2) == -1    # G: separate carrier -> w=-1 (top-form)

# ...and dust w=0 if vacuum shares the one current -> the two are mutually exclusive (G1,G2)
```

All 27 asserts pass; these are the checks that forced the §12 split and the UN reclassification.

Self-reference and pattern audit (Pass C). Pass B caught a v1.8 over-reach; Pass C catches two residual v1.9 over-statements (the §12 energy-fraction claim left unconditional, and UN stated as a local identity). This is now a *three-version* pattern — v1.0/A19 → v1.8 → v1.9 each had an over-statement caught by the next audit — and Review 2 named the meta-fact directly: **internal iteration converges to honesty, not to closure**; each declared "closure" is partly walked back by the next audit. The disciplined conclusion is *not* to attempt a v2.1 closure (which the pattern predicts a v2.2 would walk back) but to (i) conclude A28 at its in-paper ceiling, (ii) hand the crux to ZS-A29 as decidable, loop-resistant computations, and (iii) seek external adversarial review of a narrow, jargon-reduced statement — the very mechanism that produced Passes B and C. The correct one-line statement: *the rank fraction 83/121 is a closed projector fact (modulo the decidable PT denominator); the energy fraction 83/121 is conditional on a present-epoch charge–flux selection that matter dilution makes non-local; that selection — the cosmological coincidence problem in Z-Spin form — is the single load-bearing item, and it is handed, unclosed and explicitly, to ZS-A29.*

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Version History

v2.0 (June 2026): CONSISTENCY-CLOSURE pass integrating two further independent reviews of v1.9; the version at which A28 reaches its in-paper closure ceiling. **Three substantive fixes.** (i) **§12 rank-vs-energy split** (mandatory): the *rank* fraction $\mathbf{q}\Lambda = \tau_{121}(\mathbf{P}\Lambda)$ = 83/121 (a projector fact, PROVEN modulo the physical-trace denominator PT) is separated from the *energy* fraction $\mathbf{\Omega}\Lambda{,}0 = \rho\Lambda{,}0/\rho\mathrm{tot}{,}0$ = 83/121 (today's observable, DERIVED-CONDITIONAL on {VS, TFS, UN, PT, Frame}); §12 retitled and the "corpus already derives ΩΛ / coefficient not free" wording removed. The parameter-free status of the flagship number is now openly tied to the present-epoch normalization (a stake Review 2 emphasized). (ii) **UN time-dependence theorem** (§16.6, eq A28.61; verify D1–D5, E1–E3): the would-be local identity $Z_F f^2 = 2\mu_Z n$ is impossible because dust dilutes ($n \propto a^{-3}$, ZS-A20) while the top-form is constant ($\partial_\mu f = 0$); equality holds on at most one hypersurface, so UN is **retracted as a local identity** and **re-cast as a present-epoch / global charge–flux selection of Σ_0** — the cosmological coincidence problem in Z-Spin form, and the genuine bridge to B3-C (which already takes $H_0 \equiv$ the Σ_0 clock as input, so no new free parameter). Gate XB-6b renamed *Present-Epoch Boundary Charge–Flux Matching*. (iii) **Structural-dilemma theorem made explicit** (§16.3; verify G1–G2): the top-form vacuum (which forces w = −1) and single-trace protection of 38:83 are *mutually exclusive* — a separate carrier is exactly what frees the ratio — so v1.9's demotion was **forced, not optional** (this resolves the internal contradiction in v1.8 that asserted both). **Editorial fixes** (per review): "dissolves the hierarchy" → "factorizes/localizes" (§11, §15); the residual "(v1.8)" tag in §15 → "(v2.0)"; **verification record corrected** — the attached verify_zs_a28_v2_0.py now contains **27 real fail-closed asserts** (blocks A–G; the prior core-audit script had none), the cover states this load-bearing count rather than a single "111/111," and flags the inherited ≈111-check corpus regression suite as maintained separately; Part I independence flagged as a publication-format obligation (restore proofs or publish as a companion No-Go paper). **Spin-out:** the crux — the present-epoch 38:83 selection — is handed to **ZS-A29** (*A Single-Parent Measure/Form Action for Dust and Vacuum: Global Charge–Flux Matching, the Present-Epoch Selection Problem, and the 38:83 No-Go or Closure*, new §17), with a Guendelman–Nissimov–Pacheva single-measure route [27,28] and a BV–BFV [29] PT computation. **Meta-conclusion (recorded in §17 and Appendix D Pass C):** the version record shows each draft catching the previous draft's over-statement, i.e. internal iteration converges to *honesty*, not *closure*; the disciplined next step is not a v2.1 but the decidable loop-resistant computations of A29 plus external adversarial review. **Net status unchanged in kind, sharpened in honesty:** field-theory half IMPORTED-PROVEN; identification half DERIVED-CONDITIONAL; B3-D unconditional closure PARTIAL; absolute scale terminal in B3-B. New references [27–29]. (A, Q, dim Z) = (35/437, 11, 2) LOCKED. Load-bearing audit 27/27 fail-closed asserts PASS; full corpus suite (≈111) maintained separately. Consolidated from internal Z-Spin Collaboration research notes through v2.0.0. **v1.9 (June 2026):** B3-D AUDIT + CORRECTION pass — integrating two independent adversarial reviews of v1.8. **Corrected a load-bearing overclaim:** the ZS-A20 §9.3 single-trace structure protects only the *within-carrier* ratio $\rho_b:\rho_c$ = **6:32** (baryon and CDM share one Brown current, one μZ), **not** the *cross-carrier* ratio $\rho_{matter}:\rho_\Lambda$ = **38:83**, because the vacuum is a *separate* top-form whose relative normalization $(Z_F f^2)/(2\mu Z_n)$ is free (verify block C, checks D15–D17). The v1.8 status "83:38 not reabsorbable — DERIVED" is therefore **superseded → OPEN**, promotable to DERIVED-CONDITIONAL only under an explicit **Unified-Normalization** condition (gate XB-6b). **Retracted "single remaining debt":** B3-D's residual is the *set* {Vacuum-Support, Top-Form-Selection, Unified-Normalization, Physical-Trace$_{121}$}; VS does **not** imply Top-Form-Selection. **Corrected VS status** from "DERIVED-CONDITIONAL on ZS-U4" to "DERIVED-CONDITIONAL on an explicit Exhaustive-Matter-Content axiom." **Removed the symmetry-uniqueness overclaim** (§16.2): $T \propto g$ is the metric variation of the chosen Maxwell four-form action, not a consequence of "no preferred direction" alone. **Wrote the unified action explicitly** (§16.4, eqs A28.58–59), which isolates the free cross-carrier coupling rather than hiding it. **Net status:** B3-D field-theory half **IMPORTED-PROVEN**; identification half **DERIVED-CONDITIONAL** on the four-antecedent implication (A28.60); as an *unconditional* closure, **PARTIAL** (down from v1.8's claimed DERIVED-CONDITIONAL). Absolute scale remains in B3-B (OPEN). Record fixes: title lightened to lead with the projector-valued top-form result; $M_{111} \to \mathbf{M}_{121}$ typo (§16.1); cover date corrected March → June 2026; "preserved verbatim" → "summarized; incorporated by reference" (Part I). Gates XB-6a/b/c named (XB-6b most consequential). Appendix D rewritten as the v1.9 audit record; self-reference note added (the v1.8 error parallels ZS-A19 v1.0's "32 faces → dark matter" over-reach, caught only by outside review). Cross-version conflict check (Appendix B): no new conflict; the correction makes ZS-A20's single-trace scope *more* precisely respected (within-carrier only). No downstream value changes; z* untouched. (A, Q, dim Z) = (35/437, 11, 2) LOCKED. Verification 111/111 PASS (94 v1.7 + 17 v1.9 block-D). Consolidated from internal Z-Spin Collaboration research notes through v1.9.0. **v1.8 (June 2026):** B3-D first closure attempt (superseded by v1.9) — the Vacuum-Identification bridge B3-D first attacked via the projector-valued top-form (new §16; Appendix D deep-exploration record). Realized the rank-83 trace complement $\mathbf{P}_\Lambda = \mathbf{I}_{121} - \mathbf{P}_b - \mathbf{P}_c$ as the internal support of a covariant top-form (3-form A_3, $F_4 = dA_3$), importing the PROVEN top-form theorems [23–26] to *force* $\partial_\mu \rho_\Lambda = 0$, $\mathbf{T}_{\mu\nu} = -\rho_\Lambda \mathbf{g}_{\mu\nu}$, and $\mathbf{w = -1}$ (Theorem 16.2 —

the field-theory half, which *survives* in v1.9). The **absolute** normalization was correctly reassigned to B3-B (ZS-A19.NG2, ZS-A26). **Note:** v1.8's claim that the single-trace structure protects the 38:83 *cross-carrier* ratio, and its "single residual debt = Vacuum-Support Lemma" framing, were **over-stated and are corrected in v1.9** (see above). Added references [23–26]; added gate XB-6; updated the four-way B3 classification. Verification 108/108 (94 v1.7 + 14 block-D) — superseded by the v1.9 17-check block-D. **v1.7 (June 2026):** CONSISTENCY + HONESTY pass. Fixed a real inconsistency (the suite passed both $w \neq -1$ and $w \equiv -1$); retired the Li dynamical-holographic w(z) and vanilla-w/DESI checks → one consistent constant-Λ cosmology. Distinguished $c^2 = 83/121 = \Omega_{\Lambda,0}$ (today) from the de Sitter coefficient 1 ($H_\infty = \sqrt{(83/121)}H_0$). XB-3 → RETIRED-AS-INAPPLICABLE. DEMOTED "83 = 82 + 1" to a re-expression. Added the four-way B3 classification (B3-D OPEN). Verification 94/94 PASS. **v1.6 (June 2026):** GROUNDED 83 in the corpus lineage (ZS-F2→U4→A19→A20); upgraded the assignment direction to theorem-forced; XB-3 DISSOLVES; corrected ΔN_{eff} (BBN survives, always-on CMB FALSIFIED). 88/88 PASS. **v1.5 (June 2026):** REPAIRED the v1.4 unification conflation (c^2 is a Q^2-face budget, not the Q-level Condition C); resolved A19.NG1. 69/69 PASS. **v1.4 (June 2026):** Added Part II — the X–Y holographic reframe. 55/55 PASS. **v1.3 (June 2026):** Hierarchy-exponent unification (both engines re-express N = 69.16). 40/40 PASS. **v1.2 (June 2026):** Chiral-Pfaffian: ½ not forced onto ln 3; spectral route RETIRED. 27/27 PASS. **v1.1 (June 2026):** Corrected v1.0: RETRACTED non-unitary torsion; NEW Theorems A28.7/A28.8/A28.9. 21/21 PASS. **v1.0 (June 2026):** Initial release — the ZS-A27 §8.2 forward program.